

Shi Boao

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SUMMARY

Computer Science undergraduate at HKU with hands-on experience in AI systems, and end-to-end software implementation. Interested in **AI-related engineering, systems/infrastructure**, with strong programming foundations in Python/C++ and solid training in operating systems, distributed systems, networks and databases.

EDUCATION

The University of Hong Kong Sep. 2022 – Jun. 2027 (Expected)
Bachelor of Engineering, major in Computer Science Rank: **1st/140+**, GPA: 4.12/4.3
Operating Systems A+, Database Systems A+, Computer organization A, Object-oriented programming and Java A+
ICPC Silver Medalist

University of Illinois Urbana-Champaign (Semester exchange) 2025 Spring
The Grainger College of Engineering GPA: 4.0/4.0
Distributed Systems A, Communication Networks A, Embedded Systems A+

Hangzhou No.2 High School of Zhejiang Province Sep. 2019 – Jun. 2022
NOIP in Zhejiang (2018-2020): First Prize (3×)

SKILLS

Programming: Python, C/C++, Go, Java, SQL, Haskell
Tools & Stack: Pytorch, CUDA, Git, ROS, MySQL, MongoDB, Node.js, React, HTML/CSS/JavaScript, PHP
Languages: English (TOEFL 104, Speaking 24), Mandarin (Native)

PUBLICATIONS

Promptable Visual Geometry Transformers for 4D Metric Human-Scene Reconstruction
Boao Shi, Qiao Feng, Yiming Huang, Lingjie Liu
Prepare for submission to Siggraph Asia 2026

EvoR: Evolving Retrieval for Code Generation
Hongjin Su, Shuyang Jiang, Yuhang Lai, Haoyuan Wu, Boao Shi, Che Liu, Qian Liu, Tao Yu
Accepted as The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP2024)

RESEARCH EXPERIENCE

Research Intern in UPenn GRASP Lab | 3D Vision, Computer Graphics May. 2025 – Current

- Conduct research under [Lingjie Liu](#) on visual geometry transformers for human-scene reconstruction
- Design promptable region-level geometry extraction pipelines to improve metric-scale estimation by 17.6% and human mesh recovery by 6.8%
- Implemented a simple CUDA-based fast rendering module for human mesh rendering during training, achieving 10× speedup compared to PyTorch3D

Research Intern in HKU Infobodied Lab | Computer Graphics, 3D Vision May. 2024 – May. 2025

- Worked with [Yanchao Yang](#) on computer vision and embodied interaction modeling
- Built an end-to-end human motion data collection pipeline: capture, scene/object reconstruction, pose estimation and tracking
- Use RealSense camera and Orbbec camera for motion capture. Collected and organized 100+ hours of human data to support model training and evaluation
- Trained diffusion-based human interaction inpainting model

Research Intern in HKU XLang Lab | *Information Retrieval*

Dec. 2023 – Mar. 2024

- Conducted research with [Tao Yu](#) on retrieval for code generation
- Curated long-tail programming language datasets and evaluated cross-language generalization of retrieval-generation pipelines
- Contributed to the EMNLP 2024 paper *EvoR: Evolving Retrieval for Code Generation*

PROJECTS

Robot auto-aiming system | *Robotics, Computer Vision, Control Theory, Sensors*

May. 2023 – Apr. 2024

- Implemented a real-time targeting pipeline: deep-learning-based robot detection, PnP-based 6D pose estimation, and Extended-kalman-filter-based motion state estimation and tracking
- Integrated ROS-based inter-node communication and control modules for robust perception-to-actuation flow

AWARDS & TEACHING

Scholarships & Honors

- AEON Scholarship, HK\$10,000 (2025, Top 4 in Engineering, HKU)
- URFP Overseas Research Award, HK\$40,000 (2025);
- YC Cheng Scholarship, HK\$15,000 (2025)
- HKU Alumni Prize, HK\$10,000 (2024, 1 awardee in Engineering);
- Dean's Honours List, HK\$10,000 (Top 10%)
- Talent Development Scholarship, HK\$10,000 (2024)
- Young Tsun Dart Scholarship, HK\$15,000 (2023, Top 2 in Engineering, HKU)
- Soong Ching-ling Zhiyuan Scholarship, ¥80,000/year (2022–2026)

Competition Awards

- ICPC Asia East Continent Final (2023): Bronze Medal
- ICPC Hong Kong Regional (2023): Silver Medal
- NOIP in Zhejiang (2018-2020): First Prize (3x)

Teaching Experience

- HKU COMP2113 Programming II, Teaching Assistant (Sep. 2025 – Dec. 2025)
- HKU COMP2396 Object-oriented Programming, Teaching Assistant (Sep. 2024 – Dec. 2024)
- HKU ENGG1330 Programming I, Teaching Assistant (Sep. 2023 – Dec. 2023)